The Dot Product

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Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Computing dot products and angles

Let $\mathbf{u} = \langle 2, 5 \rangle$ and $\mathbf{v} = \langle -3, 4 \rangle$. Find:

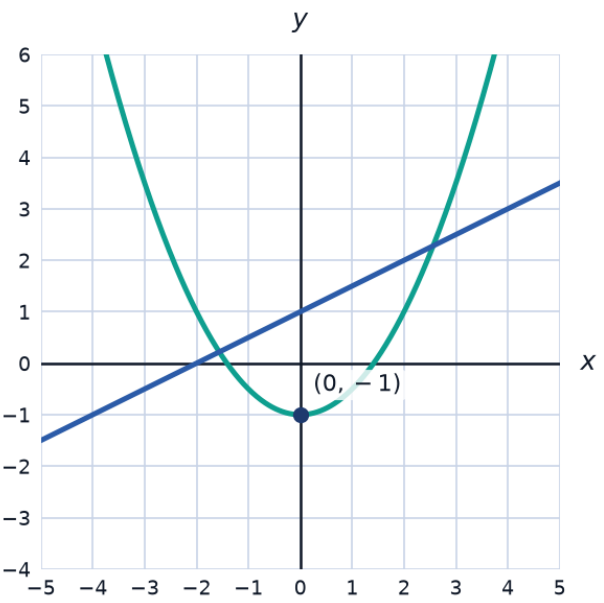
a $\mathbf{u} \cdot \mathbf{v}$

b $|\mathbf{u}|$ and $|\mathbf{v}|$

c the angle between $\mathbf{u}$ and $\mathbf{v}$, to one decimal place

2 [Foundational] • Target: Computing dot products and angles

Determine whether each pair of vectors is orthogonal:



a $\langle 4, -6 \rangle$ and $\langle 3, 2 \rangle$

b $\langle 1, 2 \rangle$ and $\langle 2, 1 \rangle$

3 [Foundational] • Target: Computing dot products and angles

Find the value of k so that $\langle k, 3 \rangle$ is orthogonal to $\langle 2, -4 \rangle$.

4 [Foundational] • Target: Projections and work

Find the vector projection of $\mathbf{u} = \langle 6, 2 \rangle$ onto $\mathbf{v} = \langle 3, 0 \rangle$.

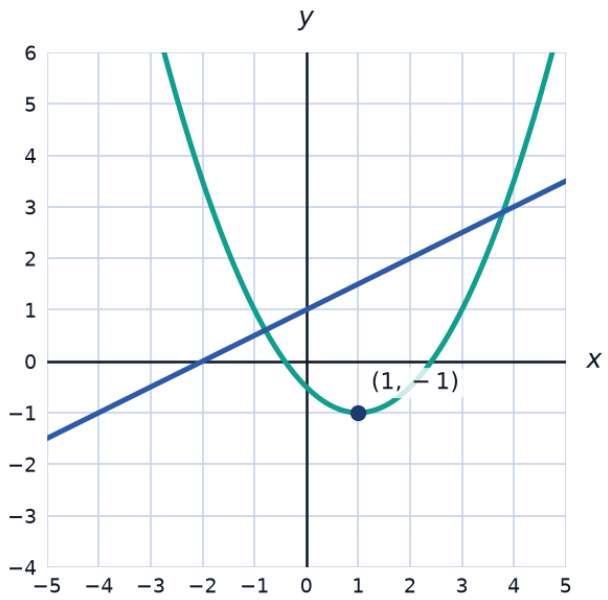
5 [Foundational] • Target: Projections and work

A force $\mathbf{F} = \langle 8, 5 \rangle$ N moves an object along the displacement $\mathbf{d} = \langle 10, 0 \rangle$ m. Find the work done.

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6 [Foundational] • Target: Projections and work

A wagon is pulled a distance of 20 m by a rope making an angle of $30°$ with the ground, with a force of magnitude 50 N. Find the work done, to one decimal place.



7 [Foundational] • Target: Properties of the dot product

Let $\mathbf{u} = \langle 3, 1\rangle$, $\mathbf{v} = \langle -2, 4\rangle$, $\mathbf{w} = \langle 1, 1\rangle$. Verify that $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$ (the distributive property).

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Properties of the dot product

Show that $\mathbf{u} \cdot \mathbf{u} = |\mathbf{u}|^2$ for $\mathbf{u} = \langle 4, -3\rangle$.

9 [Intermediate] • Target: Angle classification

Without finding the angle exactly, classify each pair of vectors as forming an acute, right, or obtuse angle, using only the sign of the dot product:

a $\langle 3, 5\rangle$ and $\langle 2, -1\rangle$

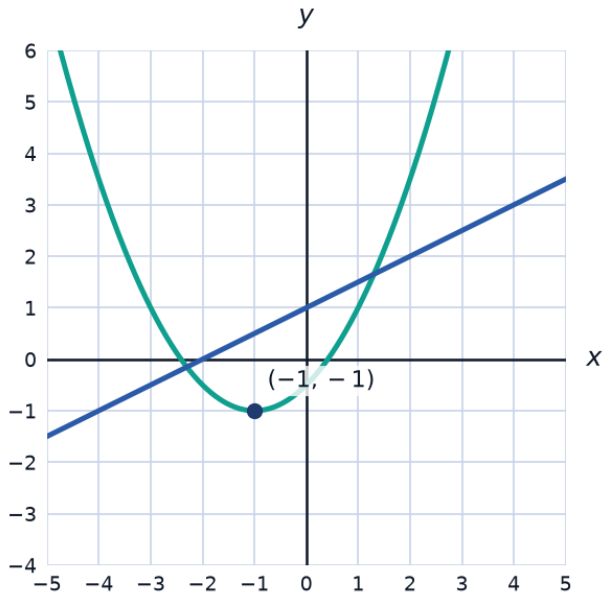
b $\langle -1, 4\rangle$ and $\langle 8, 2\rangle$

c $\langle 2, 2\rangle$ and $\langle -3, -3\rangle$

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10 [Intermediate] • Target: Scalar projection and further applications

Find the scalar projection of $\mathbf{u} = \langle 5, -2 \rangle$ onto $\mathbf{v} = \langle 3, 4 \rangle$ (the signed length $\text{comp}_{\mathbf{v}}\mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|}$), to two decimal places.



11 [Intermediate] • Target: Scalar projection and further applications

A sled is pulled by a rope at $40°$ above horizontal with a force of 120 N, moving the sled 15 m horizontally. Find the work done by the horizontal component of the force.

12 [Intermediate] • Target: Decomposing a vector into parallel and perpendicular parts

Decompose $\mathbf{u} = \langle 7, 4 \rangle$ into a component parallel to $\mathbf{v} = \langle 1, 0 \rangle$ and a component perpendicular to $\mathbf{v}$.

13 [Intermediate] • Target: Deriving the dot product angle formula

State the formula relating the dot product to the angle between two vectors, and explain why $\cos\theta = 0$ corresponds to orthogonal vectors.

14 [Intermediate] • Target: The Cauchy-Schwarz inequality (preview)

Explain why $|\mathbf{u} \cdot \mathbf{v}| \leq |\mathbf{u}||\mathbf{v}|$ must always hold, using the formula $\mathbf{u} \cdot \mathbf{v} = |\mathbf{u}||\mathbf{v}|\cos\theta$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

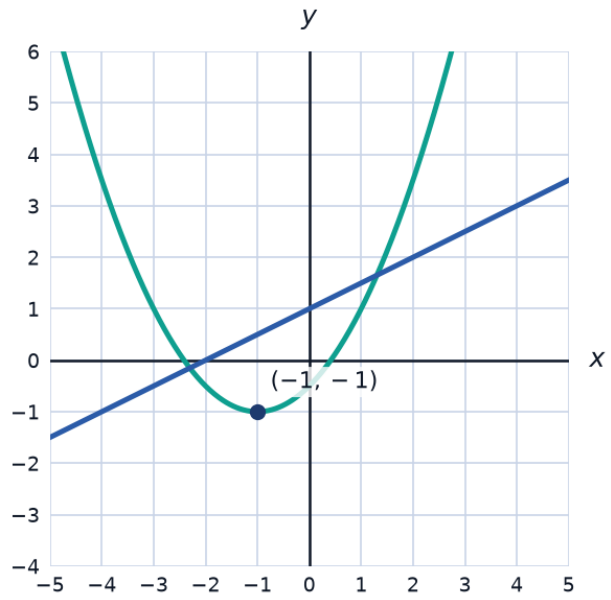
15 [Challenging] • Target: Dot products with variable components • Cross-Lesson: 1.02 Domain and Range

Find all values of k such that $\langle k, 3 \rangle$ and $\langle k, -3 \rangle$ are orthogonal.

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16 [Challenging] • Target: Dot products with variable components • Cross-Lesson: 1.03 Average Rate of Change

Find all values of k such that the angle between $\langle 1, k\rangle$ and $\langle 2, 1\rangle$ is $45°$.



17 [Challenging] • Target: A capstone dot product application • Cross-Lesson: 2.01 Right-Triangle Trigonometry

A weather vane experiences wind force $\mathbf{F} = \langle 15, -8\rangle$ N. The vane can only rotate about a pivot, so only the component of force perpendicular to the vane's current direction $\mathbf{d} = \langle 3, 4\rangle$ causes it to turn. Find the magnitude of the perpendicular (turning) component of the force, using the fact that the perpendicular component's magnitude equals $\frac{|\mathbf{F} \times \mathbf{d}|}{|\mathbf{d}|}$ in 2D (using the scalar cross product $F_x d_y - F_y d_x$), and separately confirm the parallel component using the dot product.

18 [Challenging] • Target: A capstone dot product application • Cross-Lesson: 2.02 Radian Measure and Arc Length

Two salespeople's monthly sales across three product categories are given by vectors $\mathbf{s}_1 = \langle 40, 25, 10\rangle$ and $\mathbf{s}_2 = \langle 15, 30, 45\rangle$ (units sold). Find $\mathbf{s}_1 \cdot \mathbf{s}_2$ and explain what this dot product does NOT directly tell you about their relative performance (versus what the individual sums would show).

19 [Challenging] • Target: Scalar projection and further applications • Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions

Decompose the vector $\mathbf{u} = \langle 6, 2\rangle$ into components parallel and orthogonal to $\mathbf{v} = \langle 3, 4\rangle$.

20 [Challenging] • Target: Decomposing a vector into parallel and perpendicular parts • Cross-Lesson: 2.04 Law of Sines and Law of Cosines

Consider two vectors, $\mathbf{u} = (3; k)$ and $\mathbf{v} = (-2; 4)$. If $\mathbf{u}$ is parallel to $\mathbf{v}$, find the value of the scalar parameter k.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Algebraic Definition and Properties

Find the dot product $\mathbf{u} \cdot \mathbf{v}$ for $\mathbf{u} = \langle 4, -3 \rangle$ and $\mathbf{v} = \langle 2, 5 \rangle$.

a. -7

b. 23

c. $\langle 8, -15 \rangle$

d. 1

22 [Foundational] • Target: Algebraic Definition and Properties

What is the value of $\mathbf{v} \cdot \mathbf{v}$ for any vector $\mathbf{v}$?

a. $||\mathbf{v}||^2$

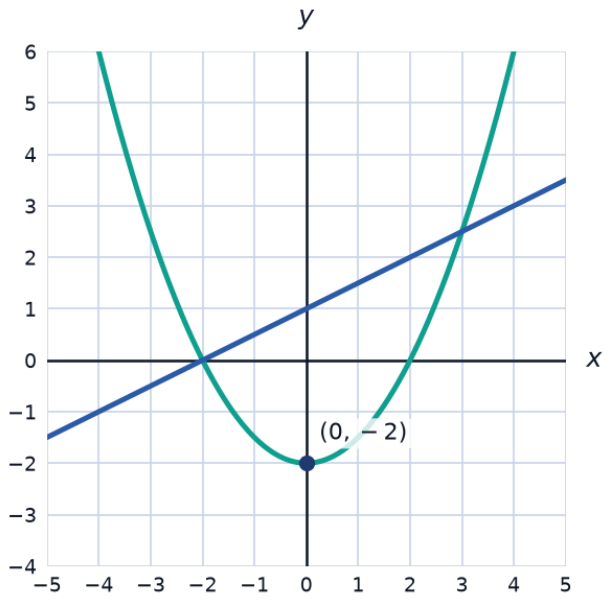
b. $||\mathbf{v}||$

c. 0

d. 1

23 [Foundational] • Target: Algebraic Definition and Properties

Which property does NOT hold for the dot product?



a. $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$ (Commutative)

b. $c(\mathbf{u} \cdot \mathbf{v}) = (c\mathbf{u}) \cdot \mathbf{v}$

c. $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$

d. $\mathbf{u} \cdot (\mathbf{v} \cdot \mathbf{w}) = (\mathbf{u} \cdot \mathbf{v}) \cdot \mathbf{w}$ (Associative)

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24 [Foundational] • Target: Geometric Definition and Angle Between Vectors

State the formula for the cosine of the angle θ between two non-zero vectors $\mathbf{u}$ and $\mathbf{v}$.

a. $\cos\theta = \frac{\mathbf{u}\cdot\mathbf{v}}{||\mathbf{u}||||\mathbf{v}||}$

b. $\cos\theta = \frac{||\mathbf{u}||||\mathbf{v}||}{\mathbf{u}\cdot\mathbf{v}}$

c. $\cos\theta = \mathbf{u}\cdot\mathbf{v}$

d. $\cos\theta = \frac{\mathbf{u}+\mathbf{v}}{||\mathbf{u}+\mathbf{v}||}$

25 [Foundational] • Target: Geometric Definition and Angle Between Vectors

Find the angle between $\mathbf{u} = \langle 1, 1\rangle$ and $\mathbf{v} = \langle 0, 2\rangle$.

a. 45°

b. 90°

c. 30°

d. 60°

26 [Foundational] • Target: Geometric Definition and Angle Between Vectors

If the dot product $\mathbf{u}\cdot\mathbf{v} < 0$, what type of angle is θ?

a. Obtuse ($90^\circ < \theta \leq 180^\circ$)

b. Acute ($0^\circ \leq \theta < 90^\circ$)

c. Right (90°)

d. Straight (180°)

27 [Foundational] • Target: Orthogonality and Parallel Vectors

Two non-zero vectors $\mathbf{u}$ and $\mathbf{v}$ are orthogonal (perpendicular) if and only if:

a. $\mathbf{u}\cdot\mathbf{v} = 0$

b. $\mathbf{u}\cdot\mathbf{v} = 1$

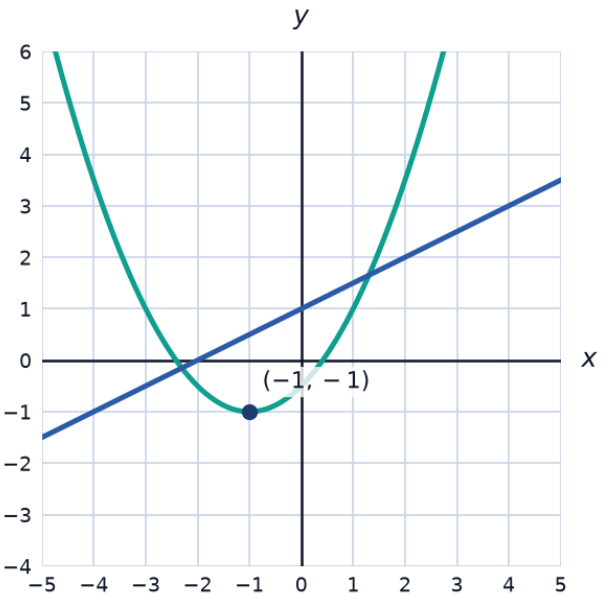
c. $\mathbf{u}\times\mathbf{v} = \mathbf{0}$

d. $||\mathbf{u}|| = ||\mathbf{v}||$

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28 [Foundational] • Target: Orthogonality and Parallel Vectors

Find k such that $\mathbf{u} = \langle 3, k \rangle$ is orthogonal to $\mathbf{v} = \langle 4, 6 \rangle$.



a. -2

b. 2

c. $-1/2$

d. 12

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Orthogonality and Parallel Vectors

Which pair of vectors is parallel?

a. $\langle 2, -3 \rangle$ and $\langle -6, 9 \rangle$

b. $\langle 2, 3 \rangle$ and $\langle 3, 2 \rangle$

c. $\langle 1, 0 \rangle$ and $\langle 0, 1 \rangle$

d. $\langle 4, 1 \rangle$ and $\langle -1, 4 \rangle$

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30 [Intermediate] • Target: Vector Projections and Work

State the formula for the vector projection of $\mathbf{u}$ onto $\mathbf{v}$ ($\text{proj}_{\mathbf{v}}\mathbf{u}$).

a. $\left(\frac{\mathbf{u}\cdot\mathbf{v}}{||\mathbf{v}||^2}\right)\mathbf{v}$

b. $\left(\frac{\mathbf{u}\cdot\mathbf{v}}{||\mathbf{u}||^2}\right)\mathbf{u}$

c. $\frac{\mathbf{u}\cdot\mathbf{v}}{||\mathbf{v}||}$

d. $(\mathbf{u}\cdot\mathbf{v})\mathbf{v}$

31 [Intermediate] • Target: Vector Projections and Work

Find $\text{proj}_{\mathbf{v}}\mathbf{u}$ for $\mathbf{u} = \langle 6, 2\rangle$ and $\mathbf{v} = \langle 4, 0\rangle$.

a. $\langle 6, 0\rangle$

b. $\langle 4, 0\rangle$

c. $\langle 0, 2\rangle$

d. $\langle 24, 0\rangle$

32 [Intermediate] • Target: Vector Projections and Work

A force $\mathbf{F} = \langle 30, 40\rangle$ N moves a cart along the displacement $\mathbf{d} = \langle 5, 0\rangle$ m. Calculate the work done.

a. 150 J

b. 200 J

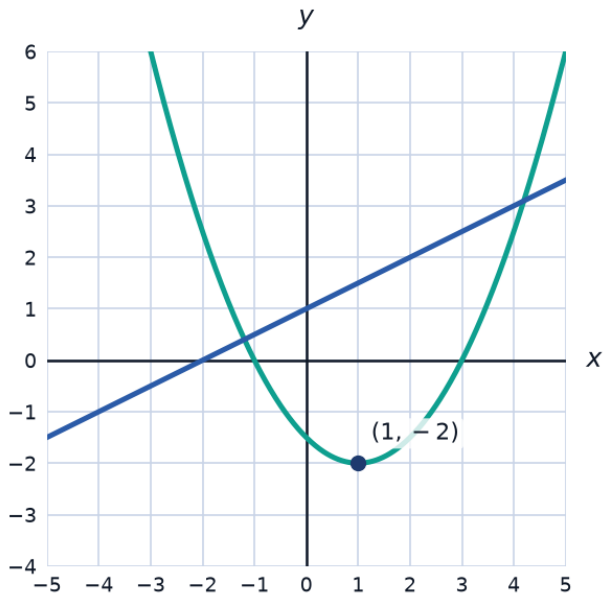
c. 350 J

d. 250 J

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33 **[Intermediate]** • **Target: Orthogonality and Parallel Vectors**

Find the vector projection of $\mathbf{u} = \langle 3, 4 \rangle$ onto $\mathbf{v} = \langle 2, 0 \rangle$.



a. $\langle 0, 3 \rangle$

b. $\langle 4, 0 \rangle$

c. $\langle 3, 0 \rangle$

d. $\langle 3, 4 \rangle$

34 **[Intermediate]** • **Target: Vector Projections and Work**

Find the vector projection of $\mathbf{u} = \langle 4, 3 \rangle$ onto $\mathbf{v} = \langle 0, 2 \rangle$.

a. $\langle 0, 3 \rangle$

b. $\langle 3, 0 \rangle$

c. $\langle 1, 3 \rangle$

d. $\langle 4, 3 \rangle$

35 **[Intermediate]** • **Target: Computing dot products and angles**

Find the vector projection of $\mathbf{u} = \langle 8, 6 \rangle$ onto $\mathbf{v} = \langle 0, 4 \rangle$.

a. $\langle 6, 0 \rangle$

b. $\langle 0, 6 \rangle$

c. $\langle 8, 6 \rangle$

d. $\langle 1, 6 \rangle$

36 **[Intermediate]** • **Target: Projections and work**

Find the vector projection of $\mathbf{u} = \langle 15, 8 \rangle$ onto $\mathbf{v} = \langle 0, 3 \rangle$.

a. $\langle 0, 8 \rangle$

b. $\langle 1, 8 \rangle$

c. $\langle 8, 0 \rangle$

d. $\langle 15, 8 \rangle$

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37 [Intermediate] • Target: Properties of the dot product

Find the vector projection of $\mathbf{u} = \langle 24, 7 \rangle$ onto $\mathbf{v} = \langle 0, 4 \rangle$.

a. $\langle 1, 7 \rangle$

b. $\langle 0, 7 \rangle$

c. $\langle 24, 7 \rangle$

d. $\langle 7, 0 \rangle$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Angle classification • Cross-Lesson: 2.02 Radian Measure and Arc Length

Find the vector projection of $\mathbf{u} = \langle 20, 21 \rangle$ onto $\mathbf{v} = \langle 4, 0 \rangle$.

a. $\langle 20, 0 \rangle$

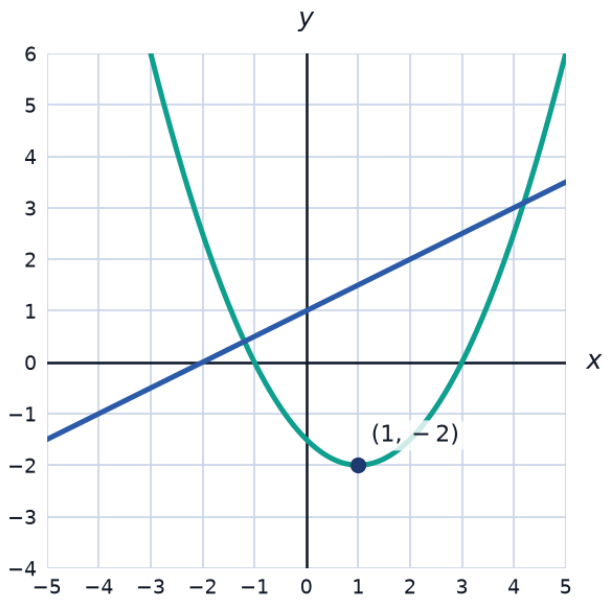
b. $\langle 21, 0 \rangle$

c. $\langle 0, 20 \rangle$

d. $\langle 20, 21 \rangle$

39 [Challenging] • Target: Scalar projection and further applications • Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions

Find the vector projection of $\mathbf{u} = \langle 21, 20 \rangle$ onto $\mathbf{v} = \langle 0, 4 \rangle$.



a. $\langle 21, 20 \rangle$

b. $\langle 20, 0 \rangle$

c. $\langle 1, 20 \rangle$

d. $\langle 0, 20 \rangle$

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40 [Challenging] • Target: Decomposing a vector into parallel and perpendicular parts • Cross-Lesson: 2.04 Law of Sines and Law of Cosines

Find the vector projection of $\mathbf{u} = \langle 6, 8 \rangle$ onto $\mathbf{v} = \langle 3, 0 \rangle$.

a. $\langle 7, 0 \rangle$

b. $\langle 6, 0 \rangle$

c. $\langle 0, 6 \rangle$

d. $\langle 6, 8 \rangle$

41 [Challenging] • Target: Deriving the dot product angle formula • Cross-Lesson: 3.01 The Unit Circle and Exact Values

Find the vector projection of $\mathbf{u} = \langle 5, 12 \rangle$ onto $\mathbf{v} = \langle 1, 0 \rangle$.

a. $\langle 0, 5 \rangle$

b. $\langle 5, 0 \rangle$

c. $\langle 6, 0 \rangle$

d. $\langle 5, 12 \rangle$

42 [Challenging] • Target: The Cauchy-Schwarz inequality (preview) • Cross-Lesson: 3.02 Fundamental Trigonometric Identities

Find the vector projection of $\mathbf{u} = \langle 12, 5 \rangle$ onto $\mathbf{v} = \langle 0, 1 \rangle$.

a. $\langle 1, 5 \rangle$

b. $\langle 0, 5 \rangle$

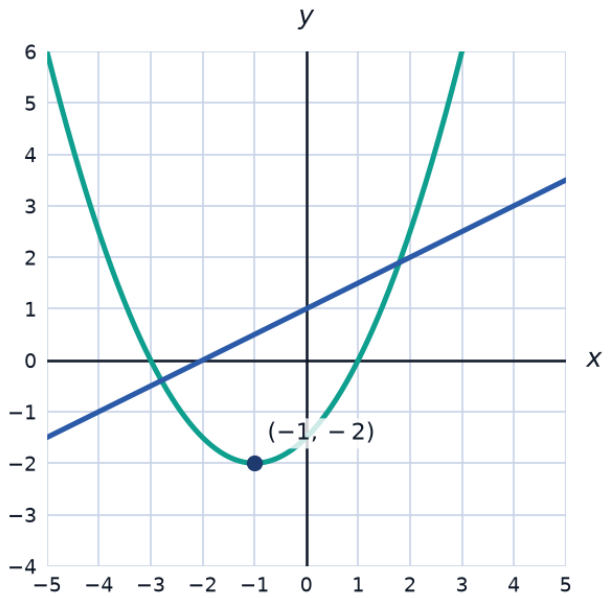
c. $\langle 5, 0 \rangle$

d. $\langle 12, 5 \rangle$

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43 [Challenging] • Target: Dot products with variable components • Cross-Lesson: 3.03 Sum and Difference Formulas

Find the vector projection of $\mathbf{u} = \langle 9, 12 \rangle$ onto $\mathbf{v} = \langle 3, 0 \rangle$.



a. $\langle 0, 9 \rangle$

b. $\langle 9, 12 \rangle$

c. $\langle 9, 0 \rangle$

d. $\langle 10, 0 \rangle$

44 [Challenging] • Target: A capstone dot product application • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations

Find the vector projection of $\mathbf{u} = \langle 12, 9 \rangle$ onto $\mathbf{v} = \langle 0, 3 \rangle$.

a. $\langle 9, 0 \rangle$

b. $\langle 0, 9 \rangle$

c. $\langle 12, 9 \rangle$

d. $\langle 1, 9 \rangle$

45 [Challenging] • Target: Algebraic Definition and Properties • Cross-Lesson: 4.01 Circles and Parabolas

Find the vector projection of $\mathbf{u} = \langle 8, 15 \rangle$ onto $\mathbf{v} = \langle 2, 0 \rangle$.

a. $\langle 8, 15 \rangle$

b. $\langle 9, 0 \rangle$

c. $\langle 8, 0 \rangle$

d. $\langle 0, 8 \rangle$

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